

**ExcaVision: An IoT-based and AI-assisted continuous monitoring system for
temporary excavation shoring**

An Application Development Project
presented to the Faculty of the
College of Computer Studies

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of the Requirements in CCS 6
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I. Introduction

ExcaVision, An IoT Based & AI Powered Continuous Monitoring of Temporary Excavation Shoring The solution addresses an important safety problem in deep excavation projects with inadequate or poorly maintained temporary excavation shoring. Due to constant environmental changes, poor stability can lead to cave-ins and structural collapse at any point in time, endangering lives and leading to damages and delays. Standard protocols usually involve periodic scheduled manual inspections and reporting, which don't adequately handle unpredictable environmental changes, such as heavy rainfall during the excavation process.

This study is important to give us an automated way to monitor shoring walls on a continuous schedule, to track movement in real time and to provide early warnings before dangerous tilt thresholds are reached.

This study gives advantages to site engineers, giving them continuous site awareness, real-time tilt alerts, and past trend data to help in making safety choices; construction contractors, who will experience less danger of excavation collapse, less financial liability, and better site safety assurance; and construction workers, who will work in a safer environment through active monitoring. Also, the system strengthens climate resilience in line with UN Sustainable Development Goal (SDG) 13.1 by mitigating climate-related excavation risks. This study is in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science, College of Computer Studies. Conducted from June to October 2026.

II. Glossary

Table 1. Table of Glossary

Term	Definitions
ExcaVision	The IoT-based and AI-assisted mobile application and hardware monitoring system under development for continuous monitoring of excavation shoring.
Sensor Node	A single hardware unit with an MPU6050 tilt sensor, placed on the shoring pilings, to get horizontal and vertical movement.
Head Sensor Node	Essentially the same unit as a sensor node, but instead of an Arduino Uno, it uses an ESP32, so that we can use its wifi feature to transmit the data collected from its previous corresponding sensor nodes to the app.
RS-485	A serial transmission protocol that allows multiple sensor nodes to be chained over a long distance from sensor node to sensor node.

TCA9548A	An I2C switch that lets the arduino nano and the ESP32 control up to 8 MPU6050 modules at once.
Early Advisory	Automated alert created by an AI trend-analysis algorithm when past tilt data is trending toward a threshold limit.
Warning Alert	A high-priority system alert is triggered if the live sensor tilt readings go above the safety threshold limit set by the site engineer.
Threshold Settings	Site engineers can set user-configurable angular limit values to ensure safe limits of operation for shoring movement.
Real-Time View	The mobile app displays a series of live tilt readings collected by the active node sensors.
History View	Historical tilt data log retrieved from the backend database can be viewed in the mobile app screen for analyzing the trend.

Site Layout	A map layout in the application that visualizes spatial arrangements and real-time status of deployed sensor nodes.
Incremental Model	The model that allows continuous creation and testing with the use of hardware, app, and AI modules.

III. User Requirements Definition

UR1: The User can use their login to safely enter into site monitoring functions.

UR2: The User will have access to the Overview dashboard with total number of monitored sites (active), total number of active sensors, and total number of recent logs from alerts.

UR3: The User will have access to a visual site map depicting the actual location and current status of all deployed sensor nodes, shall be visible to the user.

UR4: The User is able to see current tilt-data, real-time, as well as historically provided, time-series graphs per the node device selected.

UR5: The user shall be able to see push alerts on both active AI-created “Early Advisories” and urgent “Warning Alerts.”

UR6: The user shall be able to modify early advisory and warning thresholds for each sensor.

UR7: The user shall be able to manage their account settings and log out of the mobile application safely.

IV. System Requirements Specification

SR1.0: The system shall include an authentication module to protect access to applications.

SR1.1: The system shall check the user input login details against stored database accounts before giving users access to the dashboard.

SR2.0: The system shall provide a dashboard module that gives a summary of the site operational status.

SR2.1: The system shall access the backend database and give a list of active excavation sites.

SR2.2: The system shall display recent alert logs and node sensor summary data counts.

SR3.0: The system shall provide a site management module for spatial planning.

SR3.1: The system shall display site details like site name, status and number of connected sensors.

SR3.2: The system shall display an interactive map of the site layout showing the individual sensor node locations and status indicators (Normal, Early Advisory, Warning, Offline).

SR4.0: The system shall provide a sensor monitoring module for visualization of data.

SR4.1: The system shall stream real time tilt readings (X, Y axes angular displacements) from the Head Sensor Node.

SR4.2: The system shall access past sensor logs from the database and display interactive time-series trend graphs.

SR5.0: The system shall provide an alert management system and notification function.

SR5.1: The system shall support displaying of present and past alerts accessible from the backend database.

SR5.2: The system shall display detailed information about a single alert. This includes the timestamp that it was triggered at, which sensor was triggered, the recorded level of the tilt and a supporting history of patterns.

SR5.3: The system shall use Firebase Cloud Messaging (FCM) to send mobile push notifications when there are advisory or warning conditions.

SR6.0: The system shall provide a standard settings module for configuration and standard profile management.

SR6.1: The system shall allow site engineers to set and change warning thresholds for each site or for each individual sensor node.

SR6.2: The system shall allow users to modify account profile details and to log out of the session.

SR7.0: The system shall process physical sensor data transmissions thru the sensor subsystem.

SR7.1: The system shall gather raw tilt data from the MPU6050 sensors and use TCA9548A multiplexers to deal with sensor bus.

SR7.2: Sensor nodes shall send reading packets to the corresponding Head Sensor Node using an RS-485 differential bus.

SR7.3: The system shall collect sensor node packets and send telemetry data with time stamps to the mobile backend service.

SR8.0: The system shall execute automated AI trend analysis on telemetry streams.

SR8.1: The system shall pre-process and smooth the incoming tilt data to filter out temporary mechanical noise.

SR8.2: The system shall analyze the rate of change in the data given by the tilt sensors over a specified time period to identify potentially hazardous movements.

SR8.3: The system shall send an “Early Advisory” alert to the Alerts module before the raw tilt values cross the static threshold.

V. Non-functional Requirements

Product requirements

ExcaVision needs to give low latency telemetry updates to make sure that sensor reading packets sent from the corresponding Head Sensor Node are delivered to the mobile app within 3 seconds when active monitoring is on. For Android devices, the mobile application should run fine on Android 8.0 and above with a minimum of 2 GB RAM. Reliability is important. The system ensures monitoring integrity by indicating node sensor as “Offline” if the RS-485 connection fails for two consecutive cycles, in order to prevent a false sense of security, and by providing telemetry updates in the app during the specified 3 second latency period during active monitoring. The interactions around the mobile app, backend systems, and Head Sensor Node should be verified and secure by encryption. Unique color coding of the UI states Normal (Green), Early Advisory (Yellow) and Warning (Red) improves user experience.

Organizational requirements

This approach is aligned with the Incremental Model, whereby each incremental cycle consists of hardware assembly, firmware programing, mobile application design and test development, and artificial intelligent model building and test development. The firmware and mobile module sourced codes need to be checked-in into a common repository, following the standard software version controlling process (Git on Github). The programming for the embedded system hardware is carried out using embedded

C/C++, by the use of Arduino IDE, and mobile application features in cross-platform manner (using VS Code or Android Studio for building app screens).

Documentation and code shall follow the academic standards set by the College of Computer Studies.

External requirements

The system shall stick to the basic principles of the Philippine Data Privacy Act of 2012 and shall only collect operational telemetry data and user information that are necessary for the system operation without holding personal information of site construction workers. Proper credit must be given to the open source software components and firmware libraries (MIT, Apache 2.0, GPL). System documentation and user interfaces must clearly state that ExcaVision is an experimental proof-of-concept system, that it is a monitoring assistance, and that it does not replace required structural site inspection or independently promise safety for workers. If the connection to the mobile backend fails, the node sensors shall keep performing local tilt monitoring and the system shall continue automatic data transfer when the connectivity reconnects.

VI. System Architecture

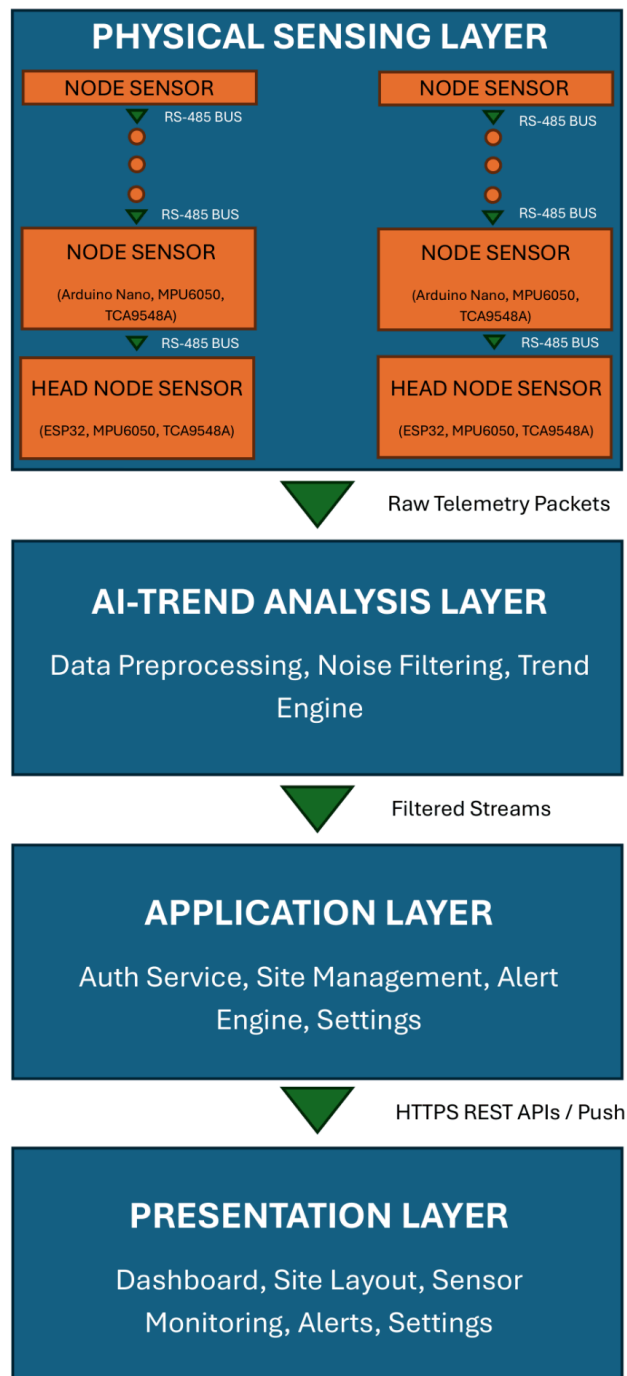


Figure 1. System Architecture Diagram

The ExcaVision system is built as a multi-layered system that separates the collection of physical hardware from the interactive mobile interfaces. The physical monitoring layer starts from each sensor node connected to the excavation pillars, where the raw tilt angle measurements are obtained by the MPU6050 sensors. These microcontrollers convert these values into serial packets of data as the TCA9548A multiplexers are actively handling sensor addressing. These packets are then transmitted over an industrial RS-485 differential bus to the Head Sensor Unit for collecting data from multiple physical nodes and transmission of the structured telemetry packets over internet connection to the mobile backend service.

The incoming telemetry data is received and the raw data is sent to the AI trend-analysis layer before being processed through the main application services. This stage handles the time-series data by smoothing it exponentially to minimize the noise resulting from the mechanical vibrations. It also constantly calculates tilt velocity to predict potential threshold branches. The filtered data along with the warning flags are then passed to the application layer that handles functions like user authentication, site layout setting, customized threshold settings and push notifications through Firebase Cloud Messaging. Finally, the presentation layer presents data that has been processed in the mobile application where site engineers can monitor live site maps, view tilt patterns through graphs, adjust tilt thresholds and receive real-time warnings. The Auth Service and Alert Engine in the Application Layer are used again in the Site Management, Sensor Monitoring and Settings modules. This allows for consistent control of access and threshold checking without repeating logic in each module.

VII. Theoretical Background

The ExcaVision project is based on fundamental physical, mathematical, and algorithmic principles in structural monitoring, hardware-based multiplexing, and analysis of time series.

Inertial Measurement & Angular Calculation

Shoring wall tilting is measured by mounting the MPU6050 6-axis Accelerometer/Gyroscope sensors on the pilings. Under static or quasi-static conditions, angular tilt (θ_x , θ_y) in relation to gravity (g) can be derived from acceleration components (A_x , A_y , A_z):

$$\theta_x = \arctan(A_y / \sqrt{(A_x^2 + A_z^2)}) \cdot (180 / \pi)$$

$$\theta_y = \arctan(-A_x / \sqrt{(A_y^2 + A_z^2)}) \cdot (180 / \pi)$$

Sensor Multiplexing & Serial Bus Communication

But how do we drive multiple, fixed addresses (we can't set the MPU6050 address) on the same local controller? The approach we take is to employ an I2C multiplexer such as the TCA9548A which enables us to sequentially activate I2C channels. To enable communication across long distances, rugged construction environments and tolerate noise across daisy-chained cabling connecting the Sensor Nodes to their respective Head Sensor Node, we employ the differential serial communication standard of RS-485.

Time-Series Smoothing & AI Trend Detection

The raw accelerometer signals are contaminated with high frequency vibration noise from nearby machinery. To prevent false alerts, an Exponential Moving Average (EMA) filter is applied to the incoming raw tilt streams (Y_t) as a preprocessing step:

$$S_t = \alpha \cdot Y_t + (1 - \alpha) \cdot S_{t-1}$$

Where S_t is the filtered tilt value at time t and α ($0 < \alpha < 1$) is the filter responsiveness parameter. The AI's lightweight trend analysis module evaluates the filtered trajectory (S_t) by a continuous slope calculator ($d\theta/dt$), over a sliding time window. If the rate of change $d\theta/dt$ is positive and continues to indicate tilt in the direction of the limit set by the engineer, the system will issue an "Early Advisory" before the threshold is reached.

VIII. System Evolution

ExcaVision is built on the following basic functional assumptions: availability of Android mobile devices with internet connection for site engineers, reliable local electricity and serial connectivity for sensor nodes, and controlled movement on a miniature excavation model.

In the future hardware evolution, miniature prototype modules might be replaced with IP67-rated weather-resistant enclosures, industrial tilt sensors, visual movement sensors, and solar-powered wireless meshing nodes for large-scale construction environments. The AI trend-analysis module could be built directly in the Head Sensor Node gateway (Edge AI) thanks to improvements in microcontroller processing power, allowing local early advisory calculation even in the absence of internet connectivity.

User needs might change to multipurpose environmental risk analysis. If more sensor types were added, for example, soil moisture detectors, pore-water pressure detectors, and rain sensors, the AI engine could associate the rate of shoring movement with rain intensity. Its flexible, multi-layer design makes it possible to update or expand the sensor hardware, artificial intelligence techniques, and mobile screens on the frontend can be upgraded or modified without changes to the primary system structure.